

SettleIQ

03 — Data Specification

Project: AI-powered settlement reconciliation and exception resolution

Purpose: Define the synthetic financial data, relationships, reconciliation rules, ground-truth generation, evidence package, and AI output structure required for the MVP.

1. Data Sources

The MVP uses three synthetic CSV datasets. Refund data is intentionally excluded from the MVP.

payments.csv → settlements.csv → fees.csv

2. payments.csv

Field	Type	Example	Purpose
payment_id	string	PAY_10001	Unique payment identifier
order_id	string	ORD_50001	Merchant order reference
payment_amount	decimal	5000.00	Amount paid
payment_date	datetime	2026-08-20 10:32	Payment timestamp
payment_status	string	SUCCESS	Payment state
customer_reference	string	CUST_8231	Customer identifier

Possible statuses: SUCCESS, FAILED, CANCELLED. Reconciliation primarily operates on successful payments.

3. settlements.csv

Field	Type	Example	Purpose
settlement_id	string	SET_70001	Unique settlement ID
payment_id	string	PAY_10001	Associated payment
settlement_amount	decimal	4850.00	Amount settled
settlement_date	datetime	2026-08-22 14:10	Settlement timestamp
settlement_status	string	SETTLED	Settlement state
settlement_reference	string	SR_98231	External settlement reference
settlement_batch_id	string	BATCH_102	Settlement batch

Possible statuses: SETTLED, PENDING, FAILED.

4. fees.csv

Field	Type	Example	Purpose
fee_id	string	FEE_90001	Unique fee ID
payment_id	string	PAY_10001	Associated payment
fee_amount	decimal	150.00	Processing fee
fee_type	string	PROCESSING_FEE	Fee category
fee_date	datetime	2026-08-20 10:33	Fee timestamp

MVP fee type: PROCESSING_FEE.

5. Primary Relationships

The primary relationship is established through payment_id:

```
payments.payment_id → settlements.payment_id
payments.payment_id → fees.payment_id
```

This allows SettleIQ to gather the payment, settlement, and fee records associated with one transaction and construct an evidence package for investigation.

6. Reconciliation Rules

Rule	Result / Logic
Payment exists	If a successful payment has no corresponding settlement → MISSING_SETTLEMENT.
Normal settlement	If settlement_amount + applicable fee_amount approximately equals payment_amount within a configured tolerance → MATCHED.
Amount mismatch	If a settlement exists but the discrepancy cannot be deterministically explained → AMOUNT_MISMATCH.
Duplicate	If multiple settlement records appear against the same payment where only one is expected → DUPLICATE.
Reference mismatch	If payment and settlement references do not line up as expected but related records may indicate a relationship → REFERENCE_MISMATCH.
Unknown / ambiguous	If no known deterministic rule can confidently classify the case → UNKNOWN.

7. Reconciliation vs. AI

The system deliberately separates deterministic reconciliation from AI investigation:

```
Raw Data → Validation → Deterministic Rules → Exception Classification → AI Investigation
```

The LLM is not used as a generic classifier over the entire database. It receives a focused evidence package only after an exception has been identified.

8. Synthetic Dataset & Ground Truth

The project will use a deterministic Python generator to create realistic but synthetic financial records. The generator will also record the ground-truth category for each generated case so that system decisions can be objectively evaluated.

Dataset	Size	Purpose
Development	50 records	Fast local development and debugging
Evaluation	200 records	Objective performance measurement
Demo	100–200 records	Realistic presentation using the same generation rules

Initial target distribution: 130 normal matches, 20 amount mismatches, 15 missing settlements, 10 duplicates, 10 reference mismatches, and 15 unknown/ambiguous cases. The distribution may be adjusted during evaluation.

9. Synthetic Data Principles

- Use deterministic Python logic rather than an LLM to generate financial ground truth.
- Introduce realistic inconsistencies instead of making every record perfectly clean.
- Include missing values, duplicate records, inconsistent references, timing differences, amount discrepancies, and missing settlements.
- Keep the ground-truth label for every injected anomaly.
- Use the same generation rules for development, evaluation, and demo datasets.

10. AI Evidence Package

For an exception, the backend gathers only relevant records before calling the AI model. A representative evidence package is:

```
{ "payment": {"payment_id": "PAY_10001", "amount": 5000}, "settlement": {"settlement_id": "SET_70001", "amount": 4850}, "fee": {"fee_id": "FEE_90001", "amount": 150, "type": "PROCESSING_FEE"}, "exception_type": "AMOUNT_MISMATCH" }
```

11. AI Output Schema

The AI should return structured output rather than unrestricted prose:

```
{ "exception_type": "AMOUNT_MISMATCH", "root_cause": "PROCESSING_FEE", "confidence": 0.96, "recommended_action": "AUTO_RESOLVE", "explanation": "The settlement is lower than the payment by exactly the recorded processing fee.", "evidence_ids": ["PAY_10001", "SET_70001", "FEE_90001"] }
```

12. Evaluation

The evaluation compares system output against the synthetic ground truth.

Metric	Meaning
Reconciliation accuracy	Correct matched/exception classification
Exception classification accuracy	Correct identification of the exception category
Auto-resolution accuracy	Percentage of automatic resolutions that are actually correct
Auto-resolution coverage	Percentage of exceptions resolved without human intervention
False auto-resolution rate	Incorrect cases that the system automatically resolved

The system should prefer safe escalation over an unsupported automatic resolution.

13. Data Scope for MVP

Included: payments, settlements, fees, five exception categories, deterministic reconciliation, evidence retrieval, AI investigation, confidence, guardrails, and evaluation ground truth.

Excluded: refunds, real Razorpay APIs, real bank integrations, real customer data, real money movement, tax reconciliation, cash forecasting, and payment execution.